

Implement dynamic data masking on specific columns (e.g., email addresses or credit card numbers) in a Synapse table to restrict data visibility based on user roles.

Created a table

The screenshot shows the Microsoft Azure Synapse Analytics interface. The left sidebar displays the 'Data' section with 'Workspace' and 'Linked' tabs. The 'Linked' tab shows 'Azure Data Lake Storage Gen2' and 'practical2synapse (Primary - data...)'. The main area shows a SQL script named 'SQL script 1' with the following code:

```
1 CREATE TABLE dbo.customers
2 (
3     customer_id INT,
4     name VARCHAR(100),
5     email VARCHAR(100) MASKED WITH (FUNCTION = 'email()'),
6     credit_card VARCHAR(20) MASKED WITH (FUNCTION = 'partial(0,"XXXX-XXXX-XXXX-",4)'),
7 )
```

The 'Results' tab shows 'No results to show' with the message 'Your query yielded no displayable results'. The 'Properties' panel on the right shows the 'General' tab with fields for 'Name' (SQL script 1), 'Description', 'Type' (.sql script), 'Size' (0 bytes), and 'Results settings per query' (First 5000 rows (default)).

Inserted values

The screenshot shows the Microsoft Azure Synapse Analytics interface. The left sidebar displays the 'Data' section with 'Workspace' and 'Linked' tabs. The 'Linked' tab shows 'Azure Data Lake Storage Gen2' and 'practical2synapse (Primary - data...)'. The main area shows a SQL script named 'SQL script 1' with the following code:

```
1 INSERT INTO dbo.customers VALUES (1, 'Tushar', 'tushar@gmail.com', '1234-5678-9012-3456');
2
3 INSERT INTO dbo.customers VALUES (2, 'ZYX', 'ZYX@yahoo.com', '9876-5432-1098-7654');
```

The 'Results' tab shows 'No results to show' with the message 'Your query yielded no displayable results'. The 'Properties' panel on the right shows the 'General' tab with fields for 'Name' (SQL script 1), 'Description', 'Type' (.sql script), 'Size' (0 bytes), and 'Results settings per query' (First 5000 rows (default)).

Created a user for testing

The screenshot shows the Microsoft Azure Synapse Analytics interface. The left sidebar displays the 'Data' section with 'Workspace' and 'Linked' tabs. The main pane shows an SQL script named 'SQL script 1' with the following code:

```
1 CREATE USER analyst_user WITHOUT LOGIN;  
2 GRANT SELECT ON dbo.customers TO analyst_user;
```

The 'Properties' pane on the right shows the script's details: Name 'SQL script 1', Type '.sql script', and Results settings per query set to 'First 5000 rows (default)'. The 'Results' pane at the bottom shows 'No results to show' with the message 'Your query yielded no displayable results'. A status bar at the bottom indicates '00:00:00 Query executed successfully.'

Output

The screenshot shows the Microsoft Azure Synapse Analytics interface with the same SQL script as above. The 'Results' pane now displays the output of the query. The 'View' dropdown is set to 'Table'. The output is a table with 4 columns: customer_id, name, email, and credit_card. The status bar at the bottom indicates '00:00:00 Query executed successfully.'

customer_id	name	email	credit_card
1	Tushar	XXXX@XXXX.com	XXXX-XXXX-XXXX-3456
2	ZYZ	ZXXX@XXXX.com	XXXX-XXXX-XXXX-7654